

Lab 0 ticket

Computers Speak Vectors — Friday, August 28

Name: _____ Language (circle): Octave / Python

New syntax (fill as you go)

I needed to...	Command / pattern (your language)
Create vector $\mathbf{x}$	
Length n	
Print 3rd Boyd entry	
Change an entry	
Scale / add vectors	
Plot	

Data: $x = (52, 51, 53, 55, 58, 60)$ °F, hours 7 a.m.–noon. Boyd $x_1 \dots x_6$. Octave: 1-based. Python: 0-based. Do **C0.1–C0.6 in order**.

C0.1 — Enter and index. Create $\mathbf{x}$. Print n , first, last, Boyd x_3 .

Expected: $n = 6$, first 52, last 60, third 53.

Yours: $n =$ _____ first=_____ last=_____ third=_____

Command that prints the third entry: _____

x_3 means (quantity + units): _____

C0.2 — Modify. Set 9 a.m. to 54, noon to 61. Print new $\mathbf{x}$.

Expected: (52, 51, 54, 55, 58, 61). **Yours:** _____

C0.3 — Scale. $\text{doubled} = 2 \times \mathbf{x}$ (use corrected $\mathbf{x}$). Print **doubled**.

Expected: (104, 102, 108, 110, 116, 122).

Assignment line: _____

Meaning of $\times 2$ here: _____

C0.4 — Add a vector. $\text{ones6} = \text{six ones}$; $\text{warmer} = \mathbf{x} + 5 \times \text{ones6}$. Print **warmer**.

Expected: (57, 56, 59, 60, 63, 66). **Yours:** _____

Why not use math-course “ $\mathbf{x}+5$ ” as vector addition? _____

C0.5 — Plot corrected $\mathbf{x}$, $2 \times \mathbf{x}$, and $-\mathbf{x}$ vs $i = 1..6$ (legend).

What did $\times(-1)$ do in the picture? _____

C0.6 — Predict, then check. Before running: Octave $\mathbf{v} = [10, 20, 30]$; $\mathbf{v}(2)$ or Python $\mathbf{v} = \text{np.array}([10, 20, 30])$; $\text{print}(\mathbf{v}[1])$.

Expected: 20. Prediction: _____ Actual: _____ Boyd index: v _____

Stuck on (optional): C0. _____